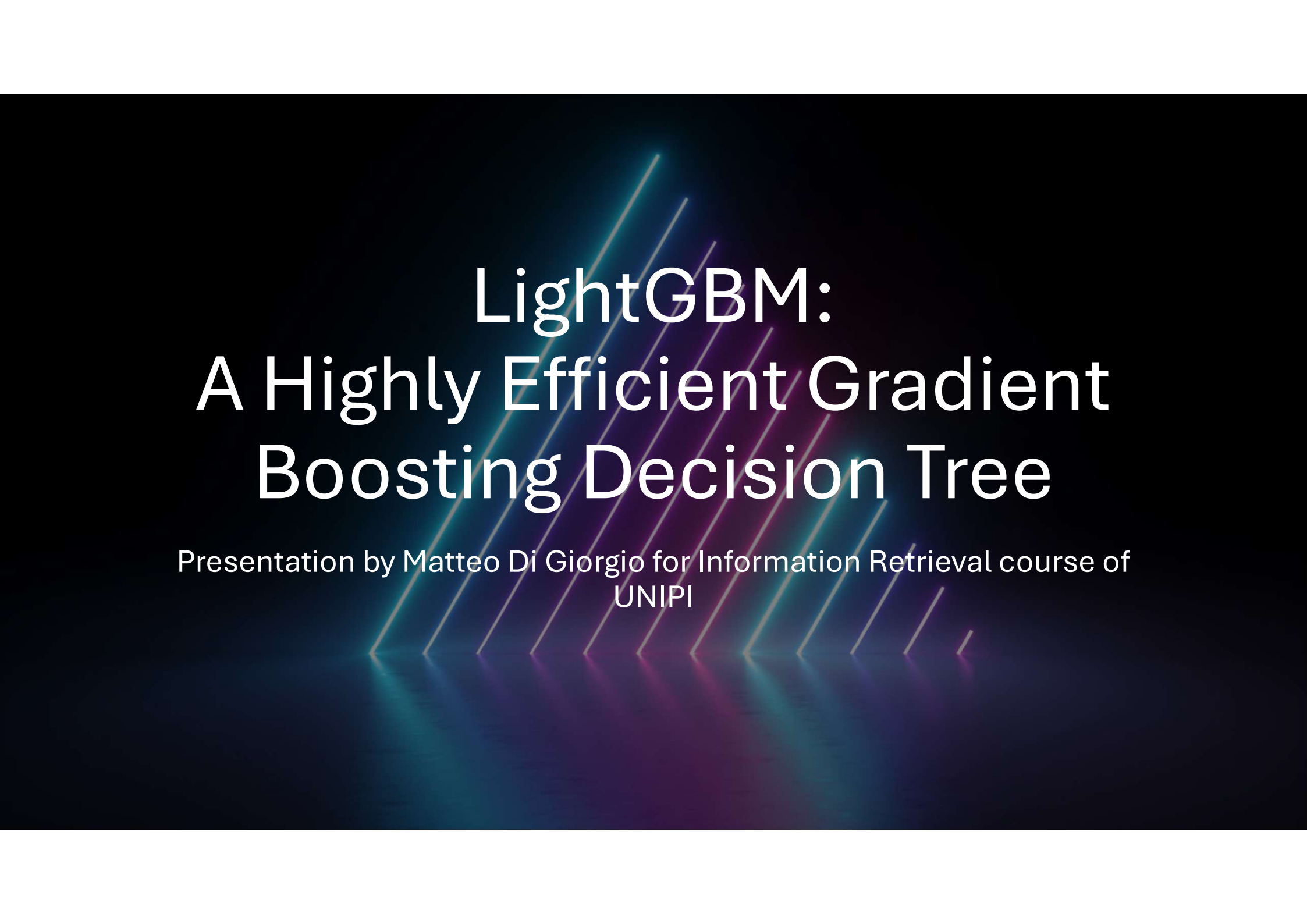




LightGBM: A Highly Efficient Gradient Boosting Decision Tree

Presentation by Matteo Di Giorgio for Information Retrieval course of
UNIFI

Introduction

LightGBM is a boosting framework developed to deal with scalability and dimensionality problems that regular Gradient Boosting Decision trees algorithm (GBDT) face

- GBDT models can get slower on large dataset or with an high volume of sparse features
- LightGBM propose two main components to deal with these issues:
 - **GOSS (Gradient-based One-Side Sampling):** reduce the number of instances based on their loss
 - **EFB (Exclusive Feature Bundling):** reduce the number of features combining the sparse ones
- LightGBM also guarantees no loss in results



Premise: Gradient Boosting Decision Trees

The Gradient Boosting Decision Tree is an algorithm that builds a sequence of decision trees, each trained to correct the errors of the previous one.

How it works:

- A initial tree that just return the mean of the target variable values is built
- The gradients of the loss function are calculated.
- A tree is built that reduce these gradients
- Second and third step are repeated until stopping conditions are met
- The final output is a weighted sum of the trees.

Main problems:

- For each possible split on each feature, all instances must be evaluated.
- This leads to a complexity of $O(n \times m)$ per split (where n = #instances, m = #features).
- On large datasets, the cost of memory and computation can become prohibitive

Gradient Based One-Side Sampling (GOSS)

GOSS is introduced to accelerate GBDT training by reducing the number of data instances used in each iteration.

- Key idea: **Not all training examples contribute equally, instances with:**
 - large gradients indicate high model error → more important.
 - small gradients indicate the model is already performing well → less critical.

GOSS selects:

- Instances with larger gradients (top a%)
- A random subset of the remaining instances (b%)

A **weight correction** is applied to the subset of the remaining instances ($w = (1-a)/b$) to maintain an unbiased estimation of information gain.

Gradient Based One-Side Sampling (GOSS)

- Given a node n and its training dataset O , the variance gain of splitting feature j at point d for this node is defined as :

$$V_{j|O}(d) = \frac{1}{n_O} \left(\frac{(\sum_{\{x_i \in O: x_{ij} \leq d\}} g_i)^2}{n_{l|O}^j(d)} + \frac{(\sum_{\{x_i \in O: x_{ij} > d\}} g_i)^2}{n_{r|O}^j(d)} \right),$$

where $n_O = \sum I[x_i \in O]$, $n_{l|O}^j(d) = \sum I[x_i \in O : x_{ij} \leq d]$ and $n_{r|O}^j(d) = \sum I[x_i \in O : x_{ij} > d]$.

- With GOSS it gets approximated to:

$$\tilde{V}_j(d) = \frac{1}{n} \left(\frac{(\sum_{x_i \in A_l} g_i + \frac{1-a}{b} \sum_{x_i \in B_l} g_i)^2}{n_l^j(d)} + \frac{(\sum_{x_i \in A_r} g_i + \frac{1-a}{b} \sum_{x_i \in B_r} g_i)^2}{n_r^j(d)} \right), \quad (1)$$

where $A_l = \{x_i \in A : x_{ij} \leq d\}$, $A_r = \{x_i \in A : x_{ij} > d\}$, $B_l = \{x_i \in B : x_{ij} \leq d\}$, $B_r = \{x_i \in B :$

- This guarantees faster computation, and its proven to not reduce accuracy

Gradient Based One-Side Sampling (GOSS)

- The following theorem (proof omitted) shows that with $n \rightarrow \infty$ the error between the true and the estimated variance tends to 0 in $O(\sqrt{n})$

Theorem 3.2 We denote the approximation error in GOSS as $\mathcal{E}(d) = |\tilde{V}_j(d) - V_j(d)|$ and $\tilde{g}_l^j(d) = \frac{\sum_{x_i \in (A \cup A^c)_l} |g_i|}{n_l^j(d)}$, $\tilde{g}_r^j(d) = \frac{\sum_{x_i \in (A \cup A^c)_r} |g_i|}{n_r^j(d)}$. With probability at least $1 - \delta$, we have

$$\mathcal{E}(d) \leq C_{a,b}^2 \ln 1/\delta \cdot \max \left\{ \frac{1}{n_l^j(d)}, \frac{1}{n_r^j(d)} \right\} + 2DC_{a,b} \sqrt{\frac{\ln 1/\delta}{n}}, \quad (2)$$

where $C_{a,b} = \frac{1-a}{\sqrt{b}} \max_{x_i \in A^c} |g_i|$, and $D = \max(\tilde{g}_l^j(d), \tilde{g}_r^j(d))$.

- It's also proven that given the generalization error in GOSS $\mathcal{E}_{gen}^{GOSS}(d) = |\tilde{V}_j(d) - V_*(d)|$,

$$\mathcal{E}_{gen}^{GOSS}(d) \leq |\tilde{V}_j(d) - V_j(d)| + |V_j(d) - V_*(d)| \triangleq \mathcal{E}_{GOSS}(d) + \mathcal{E}_{gen}(d).$$

Thus, the generalization error with GOSS will be close to that calculated by using the full data instances if the GOSS approximation is accurate

Exclusive Feature Building (EFB)

EFB addresses the high dimensionality problem in sparse datasets by reducing the number of features.

- **Many features in high-dimensional datasets are mutually exclusive**—they are never non-zero at the same time (e.g., one-hot encoded vectors).
- These features can be **safely bundled** into a single feature without information loss.

In this way, the complexity of histogram building changes from $O(\#data \times \#feature)$ to $O(\#data \times \#bundle)$, while $\#bundle \ll \#feature$. Then we can significantly speed up the training of GBDT without hurting the accuracy

Exclusive Feature Building (EFB)

- Partitioning features into a smallest number of exclusive bundles is a NP-hard problem
- Allowing conflicts in a small fraction of feature values lead to a even smaller number of feature bundles, improving computational efficiency, accuracy decreases by at most $O([(1 - \gamma)n]^{-2/3})$ where γ is the maximal conflict rate (c.r.) in each bundle
- **SELECTING THE FEATURES**
 - Sort the features by conflicts
 - Assign each feature to an existing or new bundle such that the c.r. is at most γ
 - Complexity is $O(\#feature^2)$ and its executed only once
- **BUNDLING THE FEATURES**
 - Features in the same bundle are **offset** to occupy disjoint value ranges.
 - E.g., Feature A in $\setminus[0, 10)\setminus$, Feature B in $\setminus[0, 20)\setminus \rightarrow$ B becomes $\setminus[10, 30)\setminus$
 - The merged feature retains **separability** of original features.

Experiments

The authors evaluated LightGBM on a wide variety of large-scale, real-world datasets:

- **Datasets:** MSLR-WEB10K/30K (ranking), Allstate (insurance claim prediction), KDD Cup 2010 & 2012 (advertising), Flight Delay (classification).
- **Machine:** 24 cores 256GB RAM
- **Metrics:**
 - AUC for binary classification.
 - NDCG@10 for learning-to-rank.

Findings:

- LightGBM is up to **20x faster** than XGBoost on large datasets.
- Comparable or improved accuracy, particularly for ranking tasks.
- EFB significantly reduced the number of features while maintaining fidelity.

Experiments

To validate the paper's claims, the experiments of the paper were replicated using the available datasets:

- **Datasets used:**

- MSLR-WEB30K (ranking)
- KDD 2010 (classification)
- Flight Delay (classification)

- **Hardware used:**

- 16 cores, 128GB RAM

Observations:

- Performance gains similar to those reported in the paper in term of accuracy, but the difference in hardware resulted in a difference in performance time.
- On MSLR a single fold was used due to time per connection limitations of the SSH connection

Experiments

ORIGINAL ACCURACY/NDCG@10 RESULTS:

	XGBOOST EXA	XGBOOST HIST	BASE LGBM	EFB LGBM	FULL LGBM
LETOR	0.4977	0.4982	0.5277	0.5239	0.5275
KDD10	0.7796	X	0.78735	0.7759	0.78732
FLIGHT DELAY	0.7601	0.7840	0.7847	0.7780	0.7846

REPLICATED ACCURACY/NDCG@10 RESULTS:

	XGBOOST EXA	XGBOOST HIST	BASE LGBM	EFB LGBM	FULL LGBM
LETOR	0.5605	0.5635	0.5626	0.5626	0.5626
KDD10	X	X	X	X	X
FLIGHT DELAY	0.7494	0.7819	0.7810	0.7810	0.7810

Experiments

ORIGINAL MEAN TIME PER ROUND RESULTS:

	XGBOOST EXA	XGBOOST HIST	BASE LGBM	EFB LGBM	FULL LGBM
LETOR	5.55	0.63	0.49	0.46	0.31
KDD10	108.27	X	39.85	6.33	2.85
FLIGHT DELAY	5.94	1.05	1.39	0.27	0.22

REPLICATED MEAN TIME PER ROUND RESULTS:

	XGBOOST EXA	XGBOOST HIST	BASE LGBM	EFB LGBM	FULL LGBM
LETOR	0.93	0.88	1.25	1.26	1.16
KDD10	X	X	X	X	X
FLIGHT DELAY	5.40	2.37	0.45	0.50	0.47

Conclusions

LightGBM's Contributions:

- A highly efficient GBDT framework suitable for large-scale machine learning tasks.
- GOSS and EFB bring meaningful speed-ups while preserving or improving accuracy.

Strengths:

- Handles large and sparse data efficiently.
- Combines theoretical elegance and practical performance.
- Open-source and widely adopted in industry. (Bing, DoorDash, financial sector)